

GDANSK-SWIBNO, Poland

WMO: 121550

Lat: 54.3336N

Lon: 18.9344E

Elev: 9

StdP: 101.22

Time Zone: 1.00 (EUC)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.3	-11.8	-18.0	0.8	-16.0	-14.2	1.1	-11.2	13.5	6.4	11.8	4.4	1.7	100	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.9	26.6	20.0	24.5	18.8	22.6	18.0	20.7	25.3	19.6	23.1	18.6	21.9	3.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.0	23.0	18.1	13.1	21.6	17.2	12.3	20.7	59.9	25.6	55.8	23.1	52.6	22.0	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 10.1	8.6	7.5														
(5)			DB	-15.2	30.7	6.5	1.9	-19.8	32.1	-23.7	33.2	-27.3	34.3	-32.1	35.7	
			WB	-15.5	23.2	6.3	1.4	-20.0	24.2	-23.7	25.1	-27.2	25.9	-31.8	26.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.0	-1.1	-0.8	2.2	6.8	12.0	15.2	16.9	17.2	13.0	9.3	3.6	1.1
	DBStd	7.60	6.04	5.06	4.17	3.69	3.68	3.18	2.47	2.80	2.72	3.42	3.26	3.33
	HDD10.0	1547	345	303	243	110	20	1	0	0	4	53	193	275
	HDD18.3	3823	603	536	501	347	200	104	58	57	162	279	443	533
	CDD10.0	817	0	1	1	13	82	157	215	223	94	32	0	0
	CDD18.3	52	0	0	0	0	3	11	15	21	2	0	0	0
	CDH23.3	308	0	0	0	0	24	73	78	119	13	0	0	0
	CDH26.7	58	0	0	0	0	2	14	14	25	2	0	0	0
(14) Wind	WSAvg	3.3	4.3	3.9	3.7	3.4	2.9	2.5	2.5	2.4	2.9	3.3	3.9	4.3
(15) Precipitation	PrecAvg	563	31	25	29	30	53	58	77	65	60	51	46	38
	PrecMax	705	68	44	52	73	89	140	151	143	127	118	101	64
	PrecMin	373	6	3	6	2	17	12	16	7	15	13	14	15
	PrecStd	78	15	10	12	18	23	31	34	36	32	29	21	14
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.7	12.4	15.2	21.2	26.1	28.5	28.8	29.6	25.2	19.9	11.2	10.0
		MCWB	8.5	9.3	10.3	14.5	18.7	21.5	21.8	20.6	18.2	15.7	9.6	8.9
	2%	DB	7.8	9.2	11.2	17.7	22.7	25.4	25.4	26.6	20.9	17.0	10.0	8.4
		MCWB	6.5	7.0	8.4	12.2	17.0	19.2	20.0	19.9	17.3	13.9	8.8	7.2
	5%	DB	6.6	6.6	9.4	14.6	20.1	22.7	23.2	24.3	18.9	15.3	9.0	6.9
		MCWB	5.4	5.1	7.1	10.3	15.5	17.8	18.7	18.7	15.8	13.0	7.9	5.8
	10%	DB	5.5	4.9	7.8	12.5	17.9	20.5	21.6	22.2	17.4	14.0	8.0	5.7
		MCWB	4.4	3.8	6.0	9.3	14.4	16.7	17.9	17.9	14.6	12.1	6.9	4.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.4	9.9	10.6	14.7	19.1	22.6	22.5	22.1	19.3	16.7	10.2	9.1
		MCDB	9.4	11.7	14.4	20.5	24.8	27.9	27.2	26.9	23.4	19.1	10.7	9.7
	2%	WB	6.7	7.1	8.8	12.8	17.5	19.8	20.6	20.6	17.6	14.5	9.0	7.3
		MCDB	7.6	8.8	10.8	17.4	21.9	23.8	24.7	25.1	20.6	16.4	9.7	8.2
	5%	WB	5.5	5.2	7.5	10.8	16.1	18.3	19.3	19.5	16.3	13.4	8.1	5.9
		MCDB	6.5	6.4	9.2	13.8	19.4	21.9	22.3	23.0	18.5	14.9	8.9	6.7
	10%	WB	4.4	3.8	6.0	9.5	14.8	17.2	18.3	18.4	15.1	12.4	7.1	4.8
		MCDB	5.4	4.8	7.6	12.0	17.4	20.2	21.1	21.6	16.8	13.8	7.9	5.6
(35) Mean Daily Temperature Range	5% DB	MDBR	4.3	4.6	5.7	6.7	7.1	6.9	7.2	7.9	6.6	6.3	4.2	3.8
		MCDBR	4.4	6.1	8.8	11.1	11.2	11.0	10.6	11.6	9.0	8.0	4.9	4.8
		MCWBR	4.0	4.8	6.3	6.7	6.1	5.8	6.1	5.7	5.6	5.3	4.2	4.6
	5% WB	MCDWR	4.4	5.8	8.5	10.0	9.7	9.9	9.6	9.7	8.2	7.4	4.8	4.6
		MCWBR	4.2	4.8	6.4	6.4	5.8	5.9	6.0	5.4	5.7	5.1	4.2	4.6
(40) Clear-Sky Solar Irradiance	taub		0.310	0.339	0.366	0.411	0.401	0.398	0.430	0.420	0.379	0.361	0.345	0.309
	taud		2.285	2.295	2.289	2.215	2.312	2.353	2.280	2.311	2.404	2.435	2.333	2.293
	Ebn at Noon		631	740	803	812	845	851	812	796	786	713	582	548
	Edh at Noon		59	81	101	125	119	116	123	113	90	71	57	48
(44) All-Sky Solar Radiation	RadAvg		0.49	1.16	2.40	4.18	5.40	5.65	5.08	4.37	3.10	1.59	0.60	0.35
	RadStd		0.07	0.18	0.28	0.51	0.53	0.48	0.57	0.42	0.34	0.24	0.08	0.05

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (24 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	-134	+69	N/A

Nomenclature: See separate page